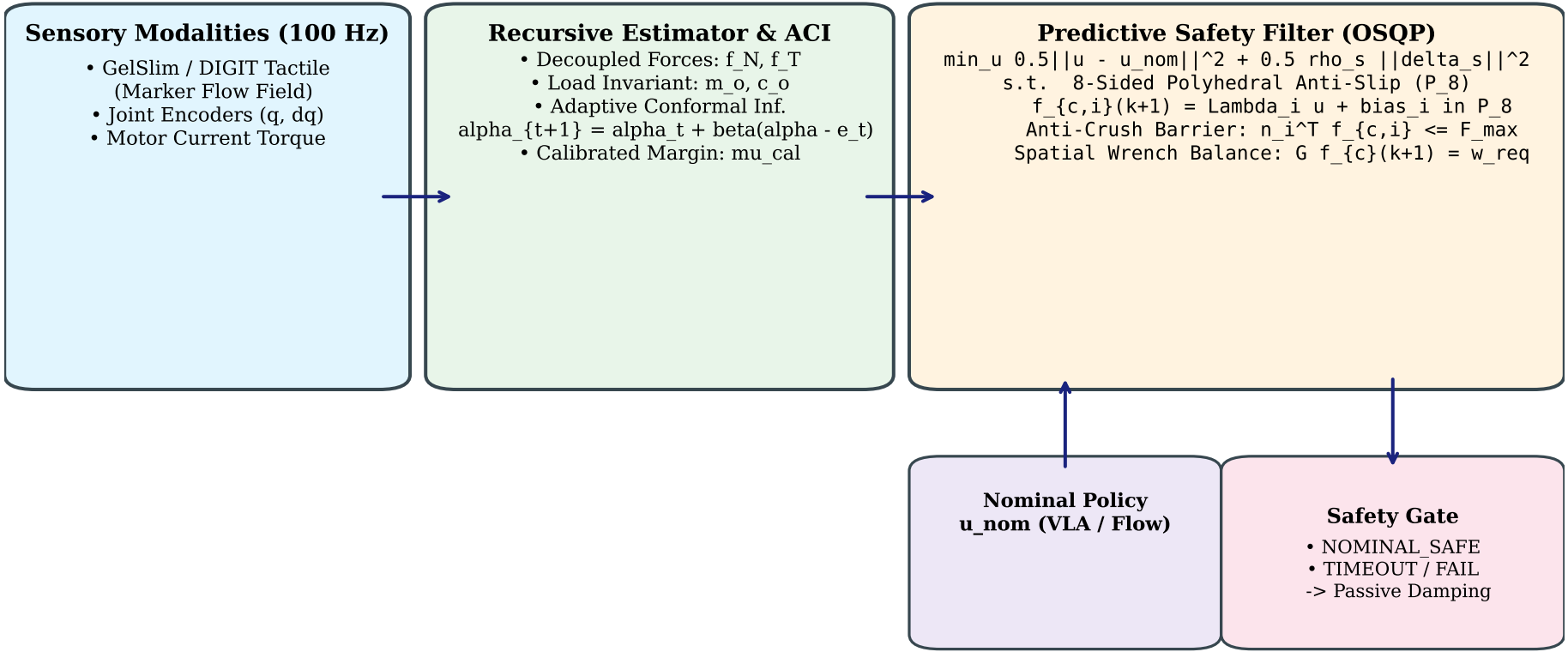
Sensory Modalities (100 Hz)
• GelSlim / DIGIT Tactile
(Marker Flow Field)
• Joint Encoders (q, dq)
• Motor Current Torque
Recursive Estimator & ACI
• Decoupled Forces: f_N, f_T
• Load Invariant: m_o, c_o
• Adaptive Conformal Inf.
alpha_{t+1} = alpha_t + beta(alpha - e_t)
• Calibrated Margin: mu_cal
Predictive Safety Filter (OSQP)
min_u 0.5||u - u_nom||^2 + 0.5 rho_s ||delta_s||^2
s.t. 8-Sided Polyhedral Anti-Slip (P_8)
f_{c,i}(k+1) = Lambda_i u + bias_i in P_8
Anti-Crush Barrier: n_i^T f_{c,i} <= F_max
Spatial Wrench Balance: G f_{c}(k+1) = w_req
Nominal Policy
u_nom (VLA / Flow)
Safety Gate
• NOMINAL_SAFE
• TIMEOUT / FAIL
-> Passive Damping